Warning Time, Alert Reach and Route Choice in Pedestrian Evacuation: An Agent-Based Simulation of a University Campus and a Nearby Town

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***Abstract*—When a sudden threat appears in a populated area, the number of people who get out of danger depends on how early they are warned, how many the warning reaches and which routes they walk. We present Panic Engine, an open agent-based simulator built on open map data, and apply it to the Amrita Vishwa Vidyapeetham campus at Amaravati and the town of Mangalagiri, 4.2 km away. The model is stated mathematically; we derive a lower bound on the number of people caught by a hazard and a formula for the discharge time of a shared route, and check both against the simulator. Four scenarios (bomb drill with a warning time, crowd stress, road closure, partial alert reach) are run on identical crowds with three routing rules and ten random crowds per setting. On the campus the share caught fell from 21.3% at a 60 s warning to 6.5% at 180 s and 0% at 300 s; in Mangalagiri it fell from 6.2% at 30 s to 0.0% at 180 s. Routing changed the number caught by about half a percentage point, but under crowding shortest-path routing evacuated only 94.0% (campus) and 79.6% (Mangalagiri) of the crowd in 30 minutes, against 100% for the other rules, and a road closure widened the gap. With 30% of people holding a phone, 28.1% of the campus crowd moved, and neighbours passing the word raised this to 85.8%. The parameters are assumptions and the model is not calibrated with field data.**

***Index Terms*—evacuation simulation, agent-based modelling, warning time, alert channels, congestion-aware routing, OpenStreetMap.**

I. Introduction

When a sudden threat appears in a populated area, such as a fire, a flood or a reported explosive device, the time between the first warning and the moment the threat takes effect is short. Three things then decide how many people get out of danger: how early they are warned, how many of them the warning reaches, and which routes they walk. None of these can be tested at full scale without danger and cost, so simulation is the practical way to compare options before a drill is planned.

General traffic simulators such as SUMO [1] and MATSim [2] can represent evacuations, but they are built for vehicle flows and take effort to adapt to one neighbourhood. A campus safety office or a town administration needs a smaller tool that runs on a laptop, uses open map data, stores every run for replay and audit, and lets a non-specialist change one setting and see the result. Panic Engine was written for this purpose. Earlier agent-based work on tsunami evacuation has shown that warning parameters change pedestrian outcomes and that a sensitivity analysis is a good way to study them [3]. We follow the same approach for a point hazard in two places in Andhra Pradesh, India: the Amrita Vishwa Vidyapeetham campus at Amaravati and the town of Mangalagiri, about 4.2 km away. The two differ strongly in road density, so they test whether a conclusion depends on the place.

The contributions are: (1) a mathematical formulation in which every equation is implemented in the code and checked against it; (2) two analytic results, a lower bound on the people caught by a hazard and a discharge-time formula for a shared route, checked against the simulator; (3) a paired design in which three routing rules run on identical stored crowds, with 95% confidence intervals for the per-crowd differences; (4) four scenarios run on both areas with ten random crowds per setting and a sensitivity study; and (5) an open code base with a dashboard, a 3-D replay and a phone-style alert screen.

II. Related Work

Helbing, Farkas and Vicsek reproduced the movement of a panicking crowd with particles that have a desired speed and repulsive interactions [4]; such force-based models suit a room or a bottleneck. Reviews of building evacuation models, such as that of Kuligowski, Peacock and Hoskins [5], show how many modelling choices exist. At the scale of a campus or a town the limits are the footpath network, the time each path takes to cross and the capacity of safe places, so we model movement on a graph without contact between individuals. Carles et al. studied how warning parameters change tsunami evacuation outcomes [3]; we vary one factor at a time in the same way for a point blast, and add alert channels and routing rules. The Bureau of Public Roads function relates link travel time to the ratio of load to capacity [6]; its constants for road traffic are 0.15 and 4, but footpaths differ, so we treat them as settings and vary them. Wardrop's principle that travellers choose the cheapest route given what others do underlies congestion-aware routing [7], and routes are found with Dijkstra's algorithm [8].

III. Study Areas and Data

Both areas use the OpenStreetMap [9] footpath network, downloaded with OSMnx [10] and held as a directed graph in NetworkX [11]. The first is a 6 km radius around the Amrita campus (16.45606° N, 80.53816° E), the second a 3 km radius around the centre of Mangalagiri (16.43182° N, 80.56881° E); the centres are 4.24 km apart (Eq. 7). Table I gives their size. Mangalagiri has 126 nodes and 26.8 km of directed road per square kilometre, the campus 61 nodes and 14.9 km, so the town is about twice as dense (Fig. 1). A synthetic grid of 80 nodes serves only as a benchmark in the crowd-stress test.

In each real area a hypothetical drill is set up in the same way. The bomb is placed at the road node nearest the map centre. Residents start at road nodes inside the alert radius from which a safe stop can be reached on foot, thinned to at most 60

start nodes. Three safe stops are chosen by a fixed rule (in each third of the compass, the node whose walking distance from the centre is closest to 1 km, or to 0.6 times the farthest reachable distance if that is smaller). They are hypothetical places, marked as such in the database, and not the official assembly points of either area.

TABLE I
THE TWO STUDY AREAS

Property	Campus	Mangalagiri
Radius of the map	6 km	3 km
Nodes	6,916	3,576
Directed road segments	18,526	9,812
Start nodes of residents	37	60
Start nodes in the blast circle	9 of 37	4 of 60
Bomb to safe stops A, B, C (m)	291, 480, 500	731, 748, 789

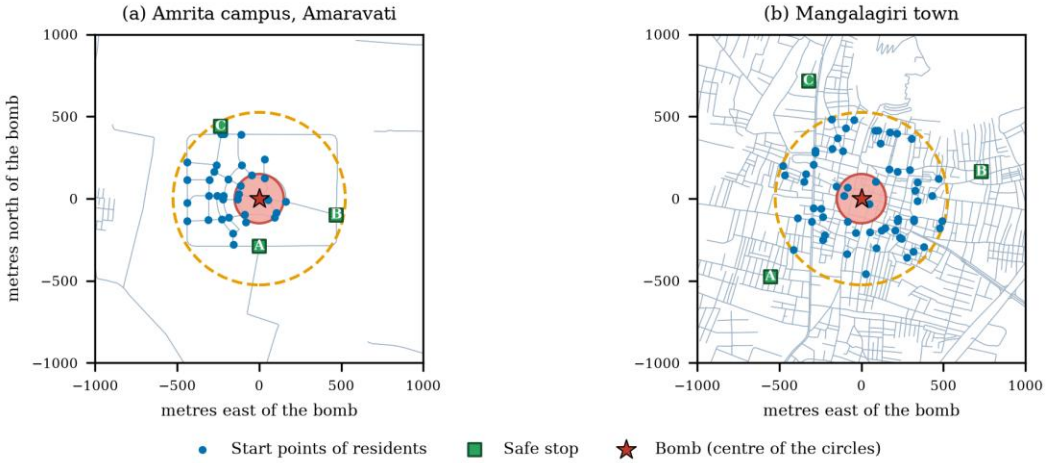


Fig. 1. The two study areas around the bomb. Blue-grey lines are footpaths and roads from OpenStreetMap, the dashed orange circle is the 525 m alert radius, the red disc the 150 m blast radius, blue dots are the start points of the residents and green squares the three safe stops.

IV. Mathematical Model

The walking network is a directed graph $G = (N, E)$; segment e has length L_e and an OpenStreetMap class c. Times are in seconds and lengths in metres. OpenStreetMap does not record how many people a path can hold, so capacity is an assumption that grows with length and with a class width factor w_c,

$$C_e = \max(2, \operatorname{round}(\rho w_c L_e)), \tag{1}$$

$\rho = 0.25$ persons per metre and w_c from 0.3 (steps) to 2.0 (primary roads), with 1.0 for residential roads; a 100 m residential road holds 25 people. With walking speed v (1.4 m/s) the free-flow time is

$$t_e^0 = \frac{L_e}{v}. \tag{2}$$

A person stepping onto e when n_e people are on it needs the Bureau of Public Roads time [6], fixed at entry from the load before that person joins, with load ratio $r_e = n_e/C_e$:

$$t_e = t_e^0\left(1 + \alpha r_e^{\beta}\right). \tag{3}$$

A segment can be entered only if $n_e < C_e$; otherwise the person waits at its tail node. Hence $r_e \le 1$ and $t_e \le (1 + \alpha)t_e^0$. We use $\alpha = 0.6$ and $\beta = 2$.

The crowd has N people; person i starts at a node o_i drawn uniformly from the start set O. With strike time τ_s and warning lead time λ the alert is sent at

$$\tau_a = \max(0, \tau_s - \lambda). \tag{4}$$

An alerted person waits a reaction delay δ_i uniform on $[0, \delta_{max}]$ and sets off at

$$d_i = \tau_a + \delta_i, \quad \mathrm{P}(\delta_i \le x) = \frac{x}{\delta_{\max}}. \tag{5}$$

The mean delay is $\delta_{max}/2$ and the variance $\delta_{max}^2/12$; with $\delta_{max} = 90$ s these are 45 s and 675 s². In the alert-channel scenario a person has a phone with probability φ and then hears the alert at $\tau_a + u_i$ (u_i uniform on [0, 15] s). Others hear nothing unless a person within ρ_n metres who already knows tells them h seconds later,

$$a_j = \min_{\mathrm{i:\ dist(i,j)} \le \rho \mathrm{n}}(a_i + h), \tag{6}$$

solved as a shortest-path problem over the people; someone nobody reaches never moves. Distances use the haversine formula on a sphere of radius $R = 6{,}371{,}000$ m,

$$\operatorname{dist}(p, q) = 2R \arcsin\sqrt{\sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos(\varphi_p)\cos(\varphi_q)\sin^2\left(\frac{\Delta\lambda}{2}\right)} \tag{7}$$

A hazard at h has blast radius R_b and alert radius $R_a = \max(3.5\,R_b, 200\ m)$, giving the blast and alert areas

$$B = \{p: dist(p, h) \le R_b\}, \quad A = \{p: dist(p, h) \le R_a\}. \tag{8}$$

At τ_s every segment with both ends in B closes and safe stops inside B are lost. Let I_s be the people not yet finished and $x_i(\tau_s)$ the position of person i; the people caught, who are the only "simulated deaths" (no injury is modelled), are

$$K = \{i \in I_s: x_i(\tau_s) \in B\}. \tag{9}$$

A route is a least-cost path found with Dijkstra's algorithm [8]. The three rules differ only in segment cost and in how a stop is chosen. Shortest path uses $c_{SP}(e) = t_e^0$; the congestion-aware rule adds the current load and the number q_e of people waiting to enter,

$$c_{\mathrm{CA}}(e) = t_e^0\left(1 + \alpha r_e^{\beta}\right) + \gamma\frac{q_e t_e^0}{C_e}, \tag{10}$$

with $\gamma = 1$. In every rule, segments in the blast area (set E_B, both ends in B) carry a penalty so nobody is sent toward the bomb when another way exists,

$$c'(e) = c(e) + \Pi \cdot 1(e \in E_B), \quad \Pi = 600\ s. \tag{11}$$

The congestion-aware and load-balanced rules also penalise crowded stops: if a_j people head for stop j of capacity K_j, the penalty is

$$\pi_j = W \frac{a_j}{K_j}, \quad W = 300\ s, \tag{12}$$

and a person at node o chooses among open stops (available and not full) with least route cost D_c:

$$j_i^* = \operatorname{argmin}_j \left[D_c(o, j) + \pi_j \right]. \tag{13}$$

Shortest path has $\pi_j = 0$. The dynamic rules re-plan every 30 s and switch only if the new score S_{new} is clearly better than the current one, S_{cur},

$$S^{\text{new}} < (1 - \theta) S^{\text{cur}}, \quad \theta = 0.15, \tag{14}$$

and at once when the next segment closes or the stop becomes full or lost. The load-balanced rule differs in who decides: a planner assigns every person a stop before the run, in departure order, by

$$j_i^* = \operatorname{argmin}_j \left[f_j, D_c(o, j) + W \frac{a_j}{K_j}, j \right], \tag{15}$$

compared lexicographically, with $f_j = 1$ if stop j is full and 0 otherwise. A stop admits 8 people per 5 s time step up to $K_j = 300$; people queue at its gate. In each step the engine closes due roads, delivers alerts, applies the strike, completes crossings, releases departures, admits people at stops and lets waiting people enter their next road.

With $\mathcal{E}$ the people admitted to a stop and t_i their admission times, we report the shares caught and evacuated and the median time from the alert $T_{med} = median(t_i - \tau_a)$ over $\mathcal{E}$:

$$P_K = \frac{100|K|}{N}, \quad P_{\mathcal{E}} = \frac{100|\mathcal{E}|}{N}. \tag{16}$$

The three rules run on the same n crowds, so rules A and B are compared by the per-crowd differences $d_k = x_k(A) - x_k(B)$, with mean $\bar{d}$, sample standard deviation s_d and 95% interval

$$\bar{d} \pm t_{0.975, n-1} \frac{s_d}{\sqrt{n}}, \tag{17}$$

$t_{0.975,\ 9} = 2.2622$ for ten crowds, computed by inverting the Student distribution.

Property 1 (free-flow bound). With D_0 the least free-flow time, Eq. (3) gives $r_e \geq 0$, so a person admitted to stop j_i satisfies

$$t_i \geq d_i + D_0(o_i, j_i). \tag{18}$$

Property 2 (blast bound). A boundary node is a node of B with a road to a node outside B; T_o is the least free-flow time from o to one. A person cannot be outside B before reaching a boundary node, so for every rule

$$|K| \geq N_{\text{LB}} = |\{i: o_i \in B,\ \delta_i + T_o > \lambda\}|. \tag{19}$$

Property 3 (expectation). For uniform delays a person starting at $o \in B$ meets the condition with probability $min\{1, max\{0, 1 - (\lambda - T_o)/\delta_{max}\}\}$, so

$$\mathrm{E}[N_{\text{LB}}] = \frac{N}{|O|} \sum_{o \in O \cap B} \min\left\{1, \max\left\{0, 1 - \frac{\lambda - T_o}{\delta_{\max}}\right\}\right\} \tag{20}$$

Property 4 (bottleneck discharge). If all N people set off together along one route, a segment holds at most C_e people who each need at least t_e, so by Little's law its throughput is at most C_e/t_e, and at full load $t_e = (1 + \alpha)t_e^0$. The narrowest segment passes about

$$q = \min_{e \in route} \frac{C_e}{(1+\alpha) t_e^0}, \quad T_k \approx T_0 + \frac{k}{q}, \tag{21}$$

with T_0 the free-flow walking time of the route and T_k the admission time of the k-th person. This is an approximation, since other segments and the stop gate also hold people back.

V. Scenarios and Parameters

Every scenario runs on both areas with the three rules on ten random crowds (seeds 1 to 10) of 250 people; each crowd is stored once and used by all three rules. *S1, bomb drill:* a 150 m bomb at the map centre goes off at $\tau_s = 420$ s and residents walk to the safe stops; λ = 60, 180, 300, 400 s on the campus and 30, 60, 120, 180, 300 s in Mangalagiri. *S2, crowd stress:* no hazard; all people (300 on the grid) leave the single start point together and the run stops at 1800 s. *S3, road closure:* as S2, but a 150 m danger zone centred on the middle of the free-flow shortest route to the nearest stop closes its roads at 300 s; people on a closed road finish it. *S4, partial alert reach:* as S1 with λ = 180 s (campus) or 60 s (Mangalagiri), but only a share φ = 100%, 60% or 30% of the crowd hears the alert on a phone, or 30% with neighbours passing the word (50 m, 20 s per hop).

Table II lists the parameters. Most are assumptions: OpenStreetMap does not contain path capacities, reaction delays or stop capacities, and we did not measure them. We vary the important ones in Section VIII and do not present absolute results as predictions. The default speed is close to the mean free walking speed of about 1.34 m/s commonly attributed to the review in [12], see also [13]; α and β differ from the road-traffic values [6].

TABLE II
MODEL PARAMETERS (ALL ASSUMED)

Parameter	Value (varied)
Walking speed v	1.4 m/s (1.0 to 1.8)
Capacity ρ; width factors w_c	0.25 persons/m; 0.3 to 2.0
BPR α, β; queue weight γ	0.6, 2; 1
Longest reaction delay δ_{max}	90 s (30 to 180 s)
Phone spread; relay ρ_n, h	15 s; 50 m, 20 s
Blast R_b; alert R_a	150 m (100, 200); max(3.5 R_b, 200 m)
Strike τ_s; lead time λ	420 s; 30 to 400 s
Penalties Π, W; θ; re-plan	600 s, 300 s; 0.15; 30 s
Stop capacity K_j; admission	300 (150); 8 per step
Time step; time limits	5 s; 2400 s (S1, S4), 1800 s (S2, S3)
Crowd size N	250 (300 grid); 150, 400 tested

VI. Software and Interface

Panic Engine is written in Python. A scenario builder sets the start points, stops, hazard, closures and alert channels, and a seeded generator draws the crowd and stores it. The timestep engine of Section IV writes every event and position snapshot to an SQLite database (eighteen tables with foreign keys and CHECK constraints), and the Streamlit dashboard, the replay and all result tables read from it. The dashboard has quick controls (routing rule, walking speed, RUN, COMPARE ALL 3, PHONE) and a sidebar for the bomb, alert channels and crowd (Fig. 2a). The deck.gl replay (Fig. 2b–d) draws the alert and blast areas, every person and extruded buildings from

21,250 cached OpenStreetMap footprints, which cover 99% of the Mangalagiri road nodes. A phone-style screen shows what one tracked person sees, and that person can be taken over to choose destination and speed. Analytics, Strategies and SQL tabs report each run and run nine integrity checks.

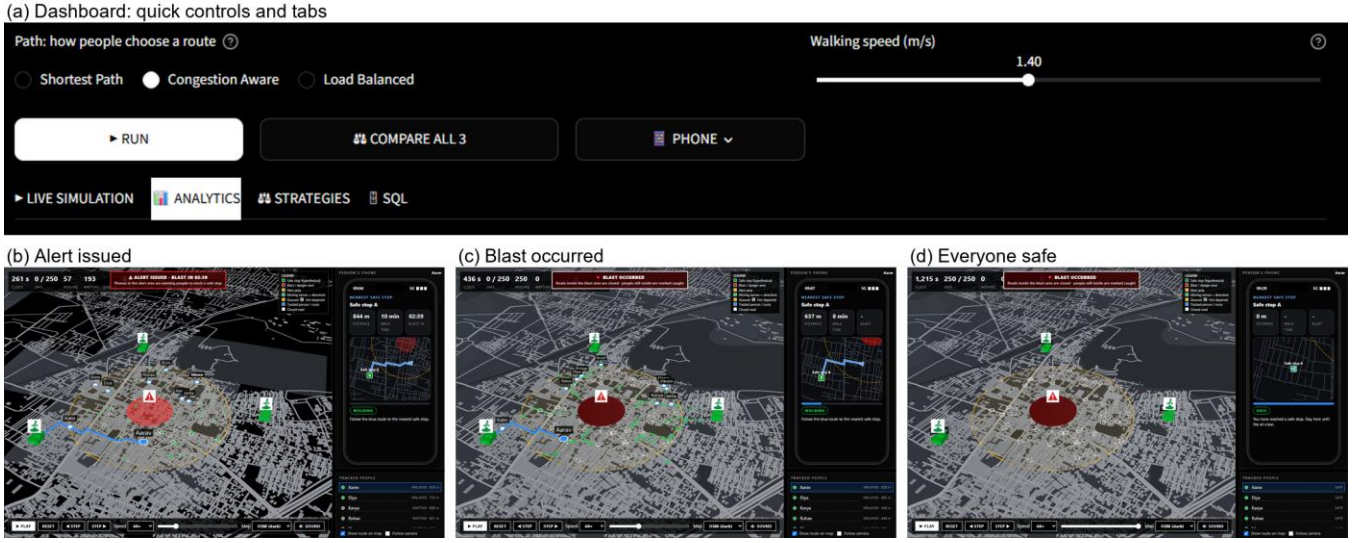


Fig. 2. The interface. (a) Quick controls and tabs of the dashboard. (b)–(d) The 3-D replay of a drill in Mangalagiri with the phone-style screen of a tracked person: the alert is issued, the blast has occurred, and everyone has reached a safe stop.

VII. Verification

We checked that the code does what Section IV says; this is verification, not validation against real events (Section IX). 82 automated tests pass, the nine integrity checks of the database pass on stored runs, and rerunning the formula checks and the paired statistics reproduced the saved results. Table III gives seven formula checks (scripts/verify_model.py).

TABLE III
NUMERICAL CHECKS OF THE FORMULAS

Check	Result
V1 (Eqs. 1, 3)	Six cases: largest code-formula difference 0.0 s; 100 m of footway, residential, tertiary, primary road hold 15, 25, 32, 50 people
V2 (Eq. 7)	Campus to Mangalagiri 4,235.66 m; flat projection differs by 6×10^{-9} (relative)
V3 (Eq. 5)	20,000 delays: mean 44.9 s (expected 45.0), variance 684 s^2 (675)
V4 (Eq. 18)	Holds for every evacuated person in all 63 runs; smallest slack 0.05 s
V5 (Eq. 19)	Caught ≥ bound in all 63 runs and rules; bound is 71% (Mangalagiri) and 53% (campus) of the simulated count
V6 (Eq. 20)	Expected vs mean realised bound. Mangalagiri 30 s: 13.9 vs 13.3; 60 s: 9.1 vs 8.3; campus 60 s: 34.1 vs 34.1; 180 s: 3.1 vs 4.0 people
V7 (Eq. 21)	Median time, predicted vs simulated: campus 1072 vs 1105 s (−3.0%); Mangalagiri 1256 vs 1305 s (−3.8%)

The blast bound is a bound, not an estimate: it ignores crowding and detours, so the simulator catches more, and that it never catches fewer (63 of 63 runs) is the check. The discharge formula is close for the median but over-predicts the share of the campus crowd evacuated in 30 minutes (99.7% against 94.0%), because the other roads and the stop gate also hold people back.

VIII. Results

A. Warning time (S1)

Table IV and Fig. 3 give the share caught at the strike. On the campus 21.3% are caught with a 60 s warning, 6.5% with 180 s and nobody with 300 s or more; in Mangalagiri the share is 6.2% at 30 s, 5.2% at 60 s, 1.5% at 120 s and 0.0% at 180 s. The most that can be caught is the share of residents who start inside the blast circle, 9 of 37 start nodes (24.3%) on the campus and 4 of 60 (6.7%) in Mangalagiri, and at the shortest lead times nearly all of them are caught. The campus is also slower to clear: the expected bound of Eq. (20) is 3.1 people at 180 s against 0.0 in Mangalagiri. Nobody in Mangalagiri reaches a stop before the strike, because the stops are 731 to 789 m away, whereas on the campus 6.6%, 16.2%, 24.0% and 35.6% are safe at 60, 180, 300 and 400 s; these shares are the same for all rules.

TABLE IV
PEOPLE CAUGHT AT THE STRIKE (%, MEAN ± SD OVER 10 CROWDS)

Area, lead time	Shortest	Congestion	Balanced
Campus, 60 s	21.3 ± 4.0	20.8 ± 4.0	20.8 ± 4.0
Campus, 180 s	6.5 ± 1.9	5.9 ± 1.8	5.8 ± 1.8
Campus, 300 s	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Campus, 400 s	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Mangalagiri, 30 s	6.2 ± 1.5	6.2 ± 1.5	6.2 ± 1.5
Mangalagiri, 60 s	5.2 ± 1.9	5.2 ± 1.8	5.2 ± 1.8
Mangalagiri, 120 s	1.5 ± 1.1	1.5 ± 1.1	1.5 ± 1.1
Mangalagiri, 180 s	0.0 ± 0.1	0.0 ± 0.1	0.0 ± 0.1
Mangalagiri, 300 s	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0

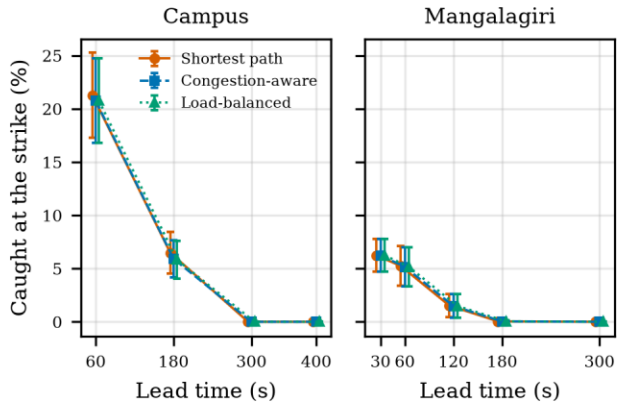


Fig. 3. People caught at the strike against the warning lead time (mean and standard deviation over ten crowds).

B. Routing rule in the bomb drills

The rules run on the same crowds, so we compare them by paired differences (Eq. 17). On the campus shortest path caught 0.52 [0.16, 0.88] percentage points more people than congestion-aware routing at a 60 s lead (mean difference and 95% interval; more in 8 of 10 crowds, fewer in none) and 0.56 [0.03, 1.09] at 180 s, about 1.4 people of 250. The standard deviation across crowds, about four points, would hide this; pairing reveals it. In Mangalagiri no interval excludes zero (60 s: 0.08 [−0.04, 0.20]). Congestion-aware routing shortened the median time from the alert on the campus by 15.0 [4.4, 25.6] s at 180 s, and the load-balanced rule did not differ from it (0.08 [−0.10, 0.26] points caught).

C. Crowd stress and road closure (S2, S3)

Without a hazard the rule matters a great deal (Table V). On the campus shortest path evacuated 94.0% of the crowd within 30 minutes against 100.0% for the other rules, with a median time from the alert of 1105 s against 805 s (congestion-aware) and 843 s (load-balanced); in Mangalagiri the shares were 79.6% and 100%. The road closure widened the gap: 82.4% against 100% on the campus and 75.6% against 97.2% in Mangalagiri. In all four scenarios shortest path evacuated fewer people in every one of the ten crowds. Property 4 explains the size: the campus route has a narrowest segment passing 0.17 people per second, so after 344 s of walking the last of 250 people arrives about 1,800 s after the alert, the end of the 30 minutes available.

TABLE V
PEOPLE EVACUATED WITHIN 30 MINUTES, NO HAZARD (%)

Scenario	Shortest	Congestion	Balanced
Campus, crowd stress	94.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Campus, road closure	82.4 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Mangalagiri, stress	79.6 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Mangalagiri, closure	75.6 ± 0.0	97.2 ± 0.0	97.2 ± 0.0
Grid, stress (300)	60.7 ± 0.2	78.1 ± 1.7	78.0 ± 1.7

D. How far the alert reaches (S4)

People who hear nothing never move, so the share evacuated is close to the phone share (Table VI): 28.1% on the campus and 28.2% in Mangalagiri with 30% phone reach. Only those inside the blast circle are caught: on the campus the share caught rose from 7.5% to 18.6%; in Mangalagiri it changed little (5.6% to 6.2%), since with a 60 s lead most people in the circle are caught even with a phone. Neighbours passing the word raised the share evacuated to 85.8% and 80.8% and cut the campus share caught to 8.6%; in Mangalagiri the relay was too slow for the 60 s lead. The rules differ by less than one point here.

TABLE VI
EFFECT OF THE SHARE OF PEOPLE WITH A PHONE (SHORTEST PATH, %)

Phone share	Campus caught	Campus evac.	Mang. caught	Mang. evac.
100%	7.5	92.5	5.6	94.4
60%	14.0	55.1	6.0	55.8
30%	18.6	28.1	6.2	28.2
30% + relay	8.6	85.8	6.2	80.8

E. Sensitivity to the assumed values

Table VII changes one factor at a time around the campus drill with a 180 s lead. Reaction delay, walking speed and blast radius move the outcome far more than the routing rule: with delays up to 30 s only 1.8% are caught against 11.8% at 180 s, and slow walking (1.0 m/s) leaves 10.0% caught against 2.9% at 1.8 m/s. The ranking of the rules is not stable: with a 200 m blast radius congestion-aware routing caught more (13.0% against 11.3%); these runs were saved without individual crowds, so no paired interval exists. In the stress test shortest path suffers most when crowding costs more or stops are smaller: 68.0% at $\alpha = 1.2$, 62.8% with half-size stops and 58.8% with 400 people (94.0% at the base values), against 100.0%, 100.0% and 100.0% for congestion-aware routing. Smarter routing helps across the range tried, but the benefit depends strongly on the assumed constants.

TABLE VII
PEOPLE CAUGHT ON THE CAMPUS, 180 S LEAD, ONE FACTOR CHANGED (%)

Change	Shortest	Congestion
Base case	6.5 ± 1.9	5.9 ± 1.8
Walking 1.0 m/s	10.0 ± 2.4	9.5 ± 2.5
Walking 1.8 m/s	2.9 ± 1.2	2.4 ± 1.3
Reaction delay ≤ 30 s	1.8 ± 1.5	1.8 ± 1.5
Reaction delay ≤ 180 s	11.8 ± 2.6	11.0 ± 2.4
Blast radius 100 m	1.0 ± 0.6	1.0 ± 0.6
Blast radius 200 m	11.3 ± 2.8	13.0 ± 2.9
400 people	7.9 ± 1.3	7.7 ± 1.4

IX. Limitations

The model has not been calibrated or validated against field data: we did not time real walks, observe a drill, or compare the output with another simulator. Verification shows that the code implements the equations, not that the equations describe real people. Capacities, admission rates, reaction delays and congestion constants are assumed (Table II), so absolute numbers should not be quoted as predictions for either place. Agents do not touch each other and a waiting person does not block the road behind, so gridlock is understated; time advances in 5 s steps; the crowd is synthetic (150 to 400 people) and the safe stops are hypothetical; the blast is a circle without shielding and no injury is modelled; people who hear no alert stay put and are counted neither as caught nor as evacuated outside the blast; OpenStreetMap may be incomplete for a new area. Our alert is a single time and reach, whereas recent work reviews how communication and social behaviour are modelled in agent-based evacuation [14] and implements the notification

procedures of emergency services in a large-scale model [15]. We study two places and one hazard type with the bomb at the map centre. Ten crowds give wide intervals for small effects, the intervals assume roughly normal differences, many comparisons are made without correction, and the sensitivity runs have no paired intervals.

X. Conclusion

Panic Engine compares warning times, alert reach and routing rules on identical crowds. We formulated the model, derived a lower bound on the people caught and a discharge formula for a shared route, and checked both with the other equations against the simulator. On the Amrita campus and in Mangalagiri, a longer warning removed the simulated deaths on the campus (300 s) and almost removed them in Mangalagiri (180 s), a shorter reaction delay and faster walking reduced them sharply, and the alert reach decided how many people moved at all. The routing rule changed the number caught by about half a point on the campus and by nothing measurable in Mangalagiri, but decided whether a crowd got out in time when roads were loaded or closed. People who walk slowly need earlier help or nearer safe places. Future work includes calibration with timed walks and a drill, a comparison with SUMO or MATSim on the same network, spill-back, the official assembly points and the real population, and a test of alert wording with real students.

Code and results: https://github.com/Ash19123/panic-engine (seeds 1 to 10 fix all random choices).

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